

PalmSource®

Conference and Expo



Extending Your Applications With Expansion Cards

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Discover how to extend your existing Palm OS® application to utilize expansion cards



Today's Agenda

- Expansion Card Basics
- Compatibility with Expansion Cards
- Extending Your Application



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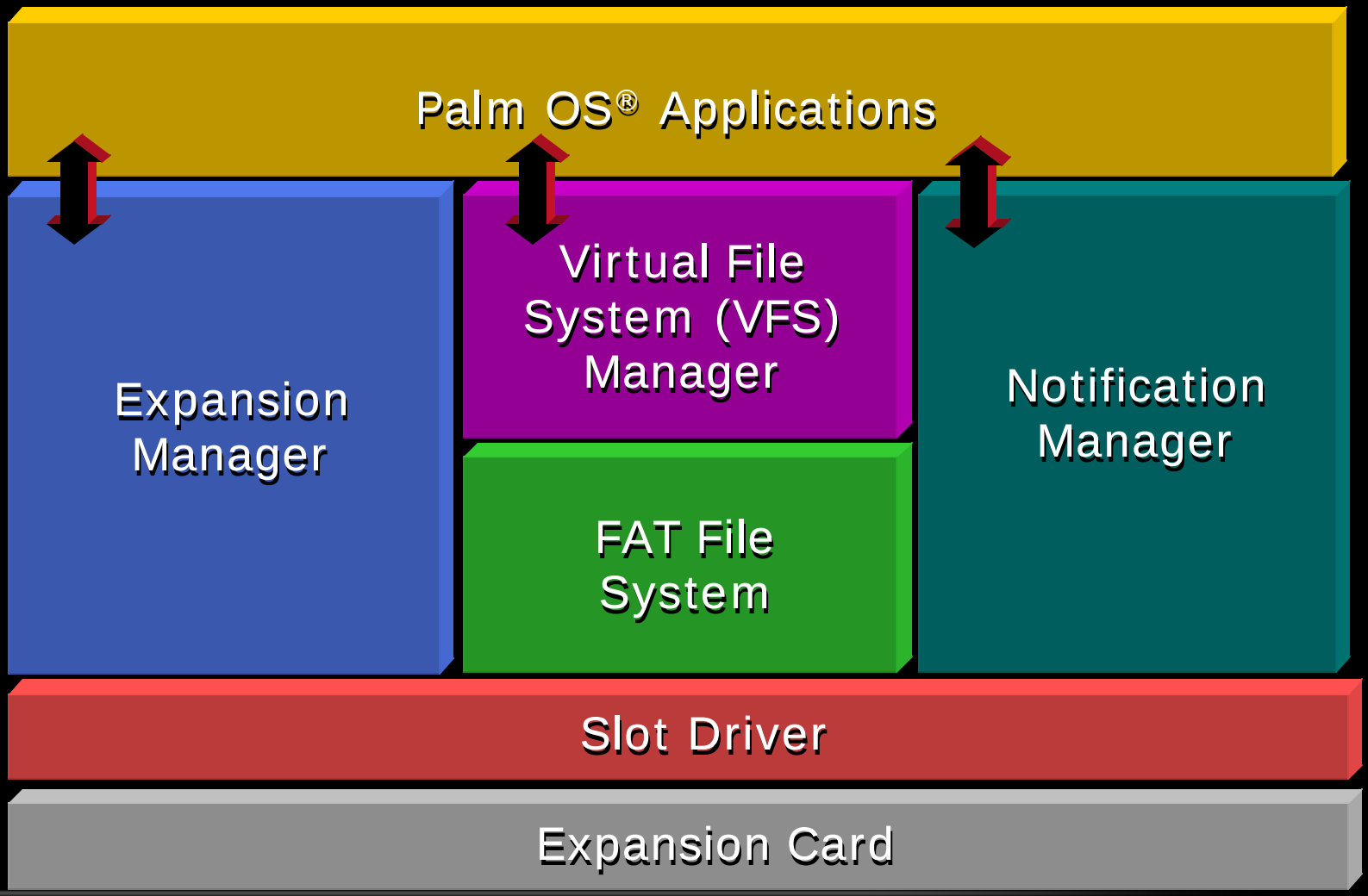
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Expansion Card Basics

- VFS Manager provides file i/o functions to Palm OS®
  - VFSFileOpen, VFSFileRead, VFSFileWrite, ...
- Notification Manager provides notifications of expansion card events
  - Card inserted, volume mounted, ...
- Expansion Manager manages it all

Functional Block Diagram



Launching from Expansion Card

- Card contents appear in Launcher
- User taps on application
- Application is copied to main memory and launched
- Application is deleted from main memory on exit





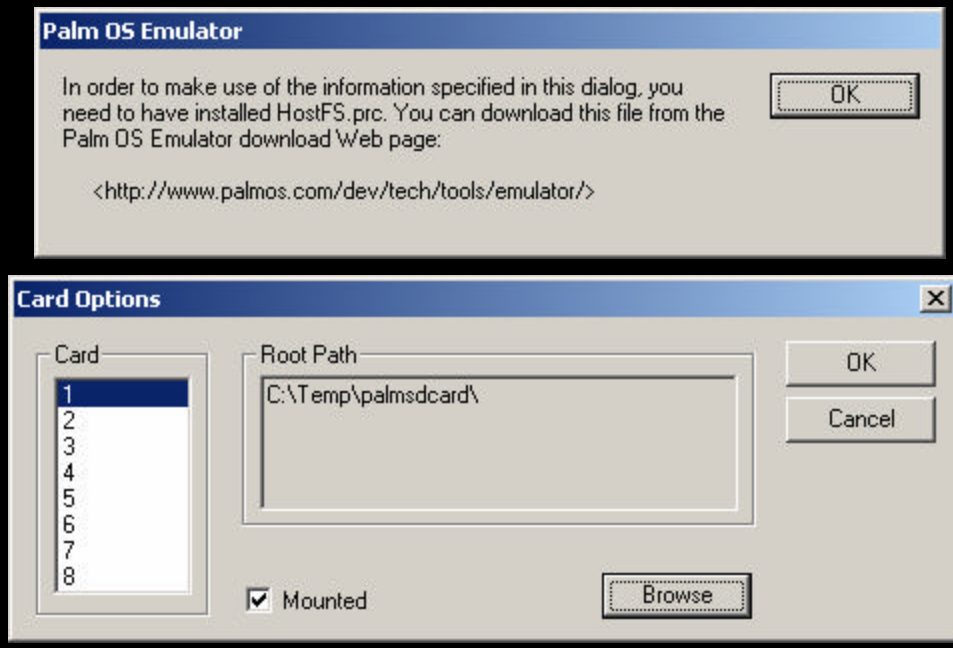
Let's Try It...

- Palm OS® Emulator (POSE®)
  - Cards supported in POSE 3.2 or later
  - Expansion Manager in Palm OS 4.0 ROM, or later
- HostFS
  - Enables file system support (cards) in POSE
  - Source code available in Knowledge Base

http://www.palmos.com/dev/tech/tools/emulator

Lab #1

- Configure POSE® to emulate an expansion card







Application Cards – Auto-Launch

- An application can be auto-launched when card is inserted
  - Name the PRC file “START.PRC” and place it in the “/PALM” folder.
  - VFS Manager handles “volume mounted” notification
  - Looks for START.PRC and launches it

Lab #2

- Configure VFS Demo application to be auto-launched from card.

Optional

- Configure VFS Demo to be launched when any card is inserted.





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Compatibility with Expansion Cards

- Test applications when launched from expansion cards
- Test applications when auto-launched from expansion cards
- Ensure application does not perform a SysUIAppSwitch to itself (Beaming)
- Test conduit when application moved to expansion card

Lab #3

- Test the VFS Demo application for compatibility with Expansion Cards



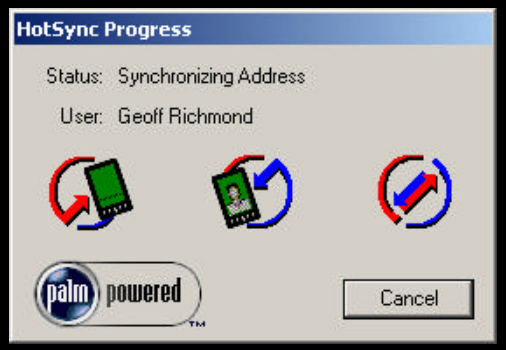
Lab #4

- Modify VFS Demo for detecting launch from card.



Lab #5

- Modify VFS Demo for HotSync® operation support when application is moved to the memory card.







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Directory Mapper

- Registry for mapping common file types (mime types) to specific directories on cards
- Different media have a different location for data
  - Images (jpg) is in “/DCIM” on Compact Flash
- Launcher registers “prc”, “pdb”, “pqa” file types for all media types
- Avoid forcing the user to navigate the card by full pathname

Lab #6

- Modify VFS Demo to search for JPEG files





Common Questions

- How do I exit the application when card is removed?
  - Handle the VolumeUnmounted notification
- What to do if application/data takes a long time to load
  - Consider loading only necessary data
  - Consider a loader application with “Please Wait” dialog



Common Problems

- Applications do a SysUiAppSwitch to re-initialize
- Applications leave orphan databases on handheld



Q&A



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